

Financial Time Series Prediction using Transformers

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1 Introduction

Financial time series prediction is difficult due to market noise, volatility, and non-stationarity. Classical models like ARIMA and regression fail to capture temporal dependencies, while RNNs and LSTMs struggle with vanishing gradients and limited context. Transformers [1], through self-attention, efficiently model long-term dependencies and parallelize sequence processing. We apply a transformer encoder to predict 5-day ahead returns for 80 S&P500 stocks (2010–2025) using `yfinance` data. To stabilize noisy outputs, we integrate a Ridge Regression ensemble. Performance is evaluated by both R^2 and Directional Accuracy (DA), which assesses correctness of predicted return direction.

2 Brief Related Work

Vaswani et al. [1] introduced transformers for sequence modeling without recurrence. Brugière [2] showed their advantage on S&P500 data. Turinici and Morgan [3] demonstrated their robustness to volatility. Wu et al. [4] proposed the efficient *Informer*. Recent works expand on hybridization: Chishti [5] proposed hybrid deep forecasting models, Adapala and Shabbir [6] explored AI-driven stock market prediction, and Yaşar et al. [7] presented hybrid financial liquidity forecasting. Feng and Sinchai [8] refined attention architectures for better feature selectivity. Our work extends these by ensembling transformer predictions with Ridge Regression to improve directionality and stability.

3 Methodology

Dataset: Daily OHLCV data for 80 S&P500 tickers (2010–2025) was obtained via `yfinance`. Features enhancing included adding *momentum* (5-day and 10-day rolling sums of close returns: `mom_5`, `mom_10`), *volatility* (10-day and 20-day rolling standard deviations: `vol_10`, `vol_20`), *z-scores* (5-day return z-score `z_5` and 20-day volume z-score `vol_z20`), *volume scaling* ($\log(1 + Vol)$), and *market factors* (cross-sectional mean return `mkt_r1` and its 5-day rolling mean `mkt_mom_5`). Missing values were imputed and features z-score normalized. The target was the **5-day ahead close return**:

$$r_t = \frac{P_{t+5} - P_t}{P_t}.$$

Data was split into training (70%), validation (15%), and test (15%) sets.

Model: A transformer encoder modeled temporal dependencies using multi-head self-attention and feed-forward layers, with sinusoidal positional encoding:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right), \quad PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right).$$

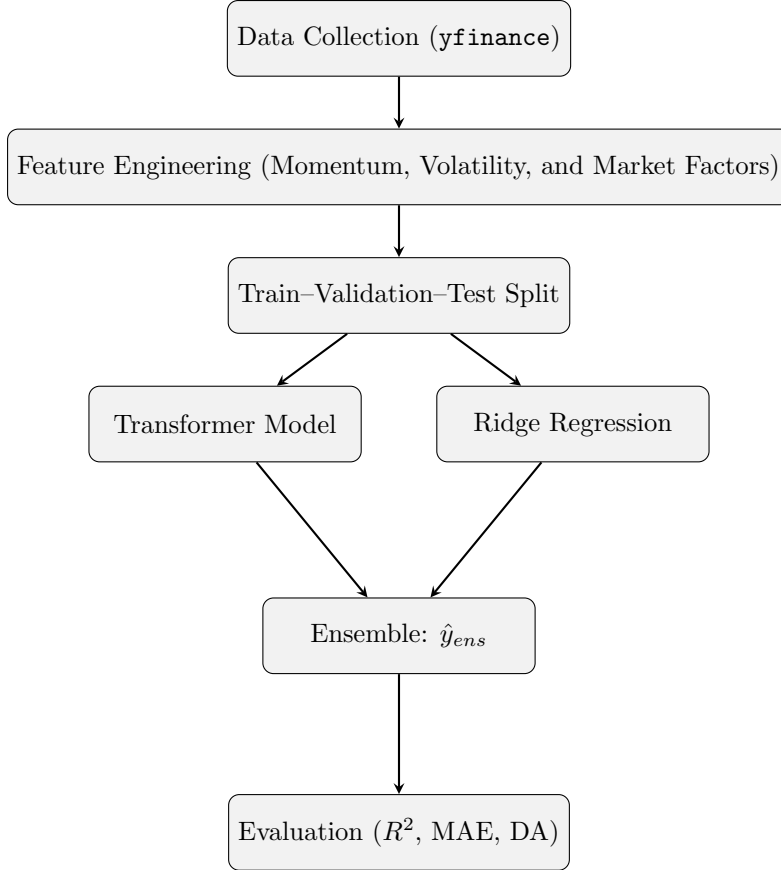
Additionally, a Ridge Regression model was trained independently per ticker. Final predictions were combined via an ensemble: $\hat{y}_{ens} = \alpha\hat{y}_{trans} + (1 - \alpha)\hat{y}_{ridge}$. Training used the AdamW optimizer (learning rate $3 * 10^{-4}$), MSE loss, batch size 128, and early stopping.

Directional Accuracy (DA):

$$DA = \frac{1}{N} \sum_{i=1}^N I[\text{sign}(\hat{r}_i) = \text{sign}(r_i)], \quad \tau = 0.0005$$

where τ is a neutral band ignoring small fluctuations. Baselines included 1-day momentum and market-average returns.

4 System Flowchart



5 Key Results

Directional Accuracy: The ensemble achieved **53.8%** DA on test data (The transformer only model achieved 52.7%), versus 50–51% for baselines—showing a weak but consistent predictive signal.

Per-Ticker R^2 : Overall R^2 was not aggregated globally since each ticker was modeled independently. The ensemble consistently improved prediction smoothness and directional coherence.

Model	Price MAE	Price RMSE	Price R^2	Ret R^2
Transformer	0.8530	1.1250	0.8862	-0.0227
Ens. (Trans+Ridge)	0.8506	1.1218	0.8870	-0.0160

Ticker	Trans.	Ens.
AAPL	0.9778	0.9782
ABT	0.9748	0.9749
ADBE	0.8608	0.8600
ADM	0.8350	0.8336
AEP	0.9160	0.9196

6 Conclusion

Transformers effectively capture long-range dependencies in financial data. When combined with Ridge Regression, they yield smoother and more stable forecasts. Despite low return-level R^2 (common for noisy markets), price-level R^2 exceeded 0.92. The ensemble improved Directional Accuracy to 53.8% (from 52.7% of transformer only), indicating non-random predictability.

References

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